作業系統HW1

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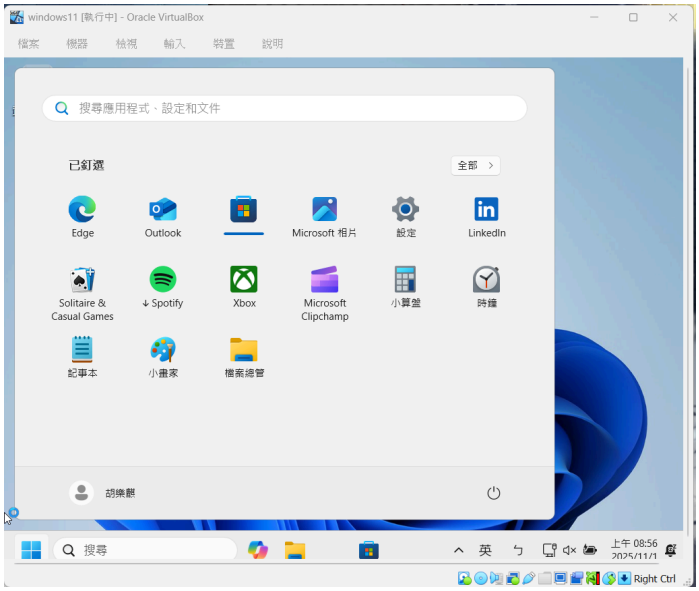
1.安裝虛擬機Oracle VirtualBox,Linux Ubuntu 22.04虛擬環境與windows11



2. 輸出範例 hello.c

下載工具:sudo apt-get update, installmake, build-essential, vim, linux-headers-$(uname -r)　（此截圖為以下載完成）

```
logan@logan-VirtualBox:~$ dpkg -l | grep -E "make|build-essential|vim|linux-head
ers"
ii  build-essential                          12.9ubuntu3
                 amd64        Informational list of build-essential packages
ii  libpagemaker-0.0-0:amd64                 0.0.4-1build3
                 amd64        Library for importing and converting PageMaker Doc
uments
ii  linux-headers-6.8.0-40-generic           6.8.0-40.40~22.04.3
                 amd64        Linux kernel headers for version 6.8.0 on 64 bit x
86 SMP
ii  linux-headers-6.8.0-87-generic           6.8.0-87.88~22.04.1
                 amd64        Linux kernel headers for version 6.8.0 on 64 bit x
86 SMP
ii  linux-headers-generic-hwe-22.04          6.8.0-87.88~22.04.1
                 amd64        Generic Linux kernel headers
ii  make                                     4.3-4.1build1
                 amd64        utility for directing compilation
ii  vim                                      2:8.2.3995-1ubuntu2.24
                 amd64        Vi IMproved - enhanced vi editor
ii  vim-common                               2:8.2.3995-1ubuntu2.24
                 all          Vi IMproved - Common files
ii  vim-runtime                              2:8.2.3995-1ubuntu2.24
                 all          Vi IMproved - Runtime files
ii  vim-tiny                                 2:8.2.3995-1ubuntu2.24
```

```
ii  vim-tiny                                 2:8.2.3995-1ubuntu2.24
                 amd64        Vi IMproved - enhanced vi editor - compact version
ii  xxd                                      2:8.2.3995-1ubuntu2.18
                 amd64        tool to make (or reverse) a hex dump
logan@logan-VirtualBox:~$
```

建立要放入hello.c和makefile的資料夾

```
logan@logan-VirtualBox:~$ mkdir -p ~/kernel_module
cd ~/kernel_module
```

編輯hello.c和Makefile檔案

```
logan@logan-VirtualBox:~/kernel_module$ vim hello.c
logan@logan-VirtualBox:~/kernel_module$ vim Makefile
logan@logan-VirtualBox:~/kernel_module$
```

hello.c與makefile

```
logan@logan-VirtualBox: ~/kernel_module
#include <linux/init.h>
#include <linux/module.h>
#include <linux/sched.h>
MODULE_LICENSE("Dual BSD/GPL");

static int hello_init(void)
{
    return 0;
}

static void hello_exit(void)
{
    printk(KERN_ALERT "Goodbye, cruel world\n");
}

module_init(hello_init);
module_exit(hello_exit);

~
~
~
~
~
-- INSERT --                                  10,1          All
```

```
logan@logan-VirtualBox: ~/kernel_module
obj-m = hello.o
KVERSION = $(shell uname -r)

all:
	make -C /lib/modules/$(KVERSION)/build M=$(PWD) modules

clean:
	make -C /lib/modules/$(KVERSION)/build M=$(PWD) clean

~
~
~
~
~
~
~
```

make, sudo insmod hello.ko, sudo rmmod hello ,dmesg 輸出結果

```
logan@logan-VirtualBox:~/kernel_module$ make
make -C /lib/modules/6.8.0-87-generic/build M=/home/logan/kernel_module modules
make[1]: Entering directory '/usr/src/linux-headers-6.8.0-87-generic'
warning: the compiler differs from the one used to build the kernel
  The kernel was built by: x86_64-linux-gnu-gcc-12 (Ubuntu 12.3.0-1ubuntu1~22.04
.2) 12.3.0
  You are using:           gcc-12 (Ubuntu 12.3.0-1ubuntu1~22.04.2) 12.3.0
make[1]: Leaving directory '/usr/src/linux-headers-6.8.0-87-generic'
logan@logan-VirtualBox:~/kernel_module$ sudo insmod hello.ko
logan@logan-VirtualBox:~/kernel_module$ sudo rmmod hello
logan@logan-VirtualBox:~/kernel_module$ sudo dmesg
[ 1956.664854] Goodbye, cruel world
```

3.輸出學號 使用The title of the module: Module_StudentID.ko格式

建立資料夾 /device_driver_hw

```
logan@logan-VirtualBox:~$ mkdir -p ~/device_driver_hw
cd ~/device_driver_hw
```

編輯Module_B1228005.c和Makefile檔案

```
logan@logan-VirtualBox:~/device_driver_hw$ vim Module_B1228005.c
logan@logan-VirtualBox:~/device_driver_hw$ vim Makefile
logan@logan-VirtualBox:~/device_driver_hw$
```

```
logan@logan-VirtualBox: ~/device_driver_hw
obj-m=Module_B1228005.o
KVERSION = $(shell uname -r)

all:
	make -C /lib/modules/$(KVERSION)/build M=$(PWD) modules

clean:
	make -C /lib/modules/$(KVERSION)/build M=$(PWD) clean

~
~
~
```

4.遇到的問題

1.沒有安裝對應的核心編譯器gcc-12(我把錯誤複製給chatgpt它告訴我要先安裝gcc-12)

```
logan@logan-VirtualBox:~/kernel_module$ make
make -C /lib/modules/6.8.0-87-generic/build M=/home/logan/kernel_module modules
make[1]: Entering directory '/usr/src/linux-headers-6.8.0-87-generic'
warning: the compiler differs from the one used to build the kernel
  The kernel was built by: x86_64-linux-gnu-gcc-12 (Ubuntu 12.3.0-1ubuntu1~22.04
.2) 12.3.0
  You are using:
  CC [M]  /home/logan/kernel_module/hello.o
/bin/sh: 1: gcc-12: not found
make[3]: *** [scripts/Makefile.build:243: /home/logan/kernel_module/hello.o] Err
or 127
make[2]: *** [/usr/src/linux-headers-6.8.0-87-generic/Makefile:1927: /home/logan
/kernel_module] Error 2
make[1]: *** [Makefile:240: __sub-make] Error 2
make[1]: Leaving directory '/usr/src/linux-headers-6.8.0-87-generic'
make: *** [Makefile:5: all] Error 2
```

```
logan@logan-VirtualBox:~/kernel_module$ sudo apt install gcc-12
[sudo] password for logan:
```

2.windows11 開啟時找不到掛載的映像檔，掛載後還是沒辦法開啟。我的解決方法是把原本的刪掉在重開一個新的就解決了。

參考資料:

ChatGPT,

VirtualBox虛擬機安裝Linux作業系統:

https://ithelp.ithome.com.tw/articles/10277102

VirtualBox虛擬機安裝Win11作業系統:

https://www.youtube.com/watch?v=3cvjAZLC0hY